TUTORIAL X

SEVERAL KINDS OF BOXES

The method of composing pages out of boxes lies at the very heart of TEX and many LATEX constructs are available to take advantage of this method of composition.

A *box* is an object that is treated by TEX as a single character. A box cannot be split and broken across lines or pages. Boxes can be moved up, down, left and right. LATEX has three types of boxes.

LR (left-right) The content of this box are typeset from left to right.

Par (paragraphs) This kind of box can contain several lines, which will be typeset in paragraph mode just like normal text. Paragraphs are put one on top of the other. Their widths are controlled by a user specified value.

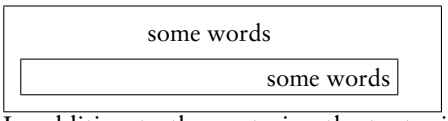
Rule A thin or thick line that is often used to separate various logical elements on the output page, such as between table rows and columns and between running titles and the main text.

X.1. LR BOXES

The usage information of four types of LR boxes are given below. The first line considers the *text* inside the curly braces as a box, with or without a frame drawn around it. For instance, `\fbox{some words}` gives some words whereas `\mbox` will do the same thing, but without the ruled frame around the text.

```
\mbox{text}
\makebox{width}{pos}{text}
\fbox{text}
\framebox{width}{pos}{text}
```

The commands in the third and fourth lines are a generalization of the other commands. They allow the user to specify the width of the box and the positioning of text inside.

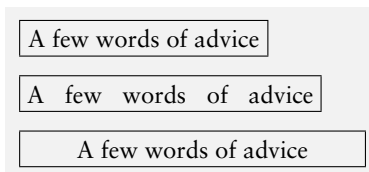



```
\makebox{5cm}{some words}    \par
\framebox{5cm}{r}{some words}
```

In addition to the centering the text with positional argument `[c]` (the default), you can position the text flush left (`[l]`). LATEX also offers you an `[s]` specifier that will stretch your text from the left margin to the right margin of the box provided it contains some stretchable space. The inter-word space is also stretchable and shrinkable to a certain extent.

With LATEX, the above box commands with arguments for specifying the dimensions of the box allow you to make use of four special length parameters: `\width`, `\height`,

\depth and \totalheight. They specify the natural size of the text, where \totalheight is the sum of the \height and \depth.




```
\framebox{A few words of advice}\\[6pt]
\framebox[5cm][s]{A few words of advice}\\[6pt]
\framebox{1.5\width}{A few words of advice}
```

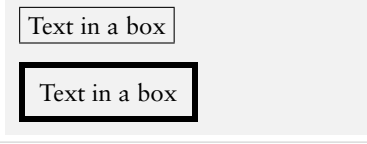
As seen in the margin of the current line, boxes with zero width can be used to make text stick out in the margin. This effect was produced by beginning the paragraph as follows:

```
\makebox{0mm}{r}{$\Leftrightarrow$}
  As seen in the margin of the \dots
```

The appearance of frameboxes can be controlled by two style parameters.

\fboxrule The width of the lines comprising the box produced with the command \fbox or \framebox. The default value in all standard classes is 0.4pt.

\fboxsep The space left between the edge of the box and its contents by \fbox or \framebox. The default value in all standard classes is 3pt.




```
\fbox{Text in a box}
\setlength\fboxrule{2pt}\setlength\fboxsep{2mm}
\fbox{Text in a box}
```

Another interesting possibility is to raise or lower boxes. This can be achieved by the very powerful \raisebox command, which has two obligatory and two optional parameters, defined as follows:

\raisebox{*lift*}{*depth*}{*height*}{*contents*}

An example of lowered and elevated text boxes is given below.

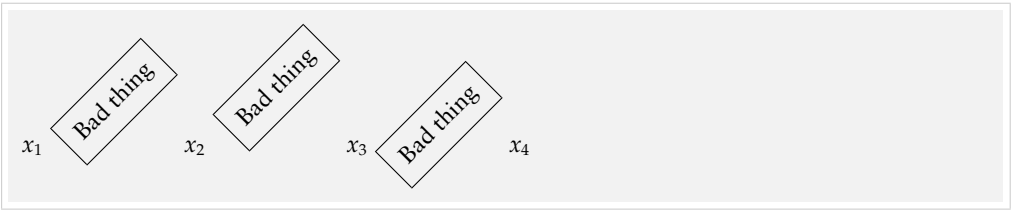



```
baseline \raisebox{1ex}{upward} baseline
      \raisebox{-1ex}{downward} baseline
```

As with \makebox and \framebox the LaTeX implementation of \raisebox offers you the use of the lengths \height, \depth, \totalheight and \width in the first three arguments. Thus, to pretend that a box extends only 90% of its actual height above the baseline you could write:

```
\raisebox{0pt}{0.9\height}{text}
```

or to rotate a box around its lower left corner (instead of its reference point lying on the baseline), you could raise it by its \depth first, e.g.:




```
$x_1$ \doturn{\fbox{Bad thing}}\\
$x_2$ \doturn{\raisebox{\depth}\\
              {\fbox{Bad thing}}}\\
$x_3$ \doturn{\raisebox{-\height}\\
              {\fbox{Bad thing}}} $x_4$
```

X.2. PARAGRAPH BOXES

Paragraph boxes are constructed using the \parbox command or minipage environment. The text material is typeset in paragraph mode inside a box of width *width*. The vertical positioning of the box with respect to the text baseline is controlled by the one-letter optional parameter *pos* ([c], [t], and [b]).

The usage for \parbox command is,

```
\parbox{pos}{width}{text}
```

whereas that of the **minipage** environment will be:

```
\begin{minipage}{pos}{width}
... here goes the text matter ...
\end{minipage}
```

The center position is the default as shown by the next example. You can also observe that LaTeX might produce wide inter-word spaces if the measure is incredibly small.

This is the contents of the left-most parbox.

CURRENT LINE

This is the right-most parbox. Note that the typeset text looks sloppy because LaTeX cannot nicely balance the material in these narrow columns.

The code for generating these three \parbox's in a row is given below:

```
\parbox{.3\bs linewidth}
 {This is the contents of the left-most parbox.} \hfill CURRENT LINE \hfill
\parbox{.3\bs linewidth}{This is the right-most parbox. Note that the typeset
 text looks sloppy because \LaTeX{} cannot nicely balance the material in
 these narrow columns.}
```